Teachers' Digital Literacy in Computer Technology Integration: Comparative Insights, AI Implications, and Practical Implementation Pathways

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Abstract

The rapid advancement of digital technologies, particularly driven by artificial intelligence (AI), has made teacher digital literacy a key priority in global education reform. While much existing research focuses primarily on technical proficiency, this study adopts a broader perspective, investigating not only technical skills but also ethical awareness, pedagogical flexibility, and contextual barriers in AI integration. Using a comparative qualitative research design, the study incorporates semi-structured interviews with 26 teachers from central China, spanning urban (n=8) and rural (n=8) school teachers, university professors (n=5), and vocational school educators (n=5). These insights are juxtaposed with international research to identify transferable strategies and context-specific adaptations. The data were analyzed thematically and supplemented with case studies from both high-resource and low-resource schools. The findings reveal significant rural–urban disparities in AI access and proficiency, with rural teachers encountering substantial infrastructural limitations and heightened concerns about AI replacing human educators, while urban teachers highlighted challenges related to data privacy and algorithmic bias. University professors exhibited the highest level of AI literacy but expressed concerns regarding the integrity of

AI-assisted assessments. The case studies emphasize that both high- and low-resource contexts can successfully integrate AI through tailored professional development and collaborative technology design. This study expands the conceptualization of teacher digital literacy in the AI era by incorporating ethical and identity dimensions and presents practical, equity-oriented pathways for educational policymakers and practitioners. The findings highlight the critical need for differentiated AI training, infrastructure investment, and teacher involvement in AI co-design to promote an inclusive and sustainable digital transformation in education.